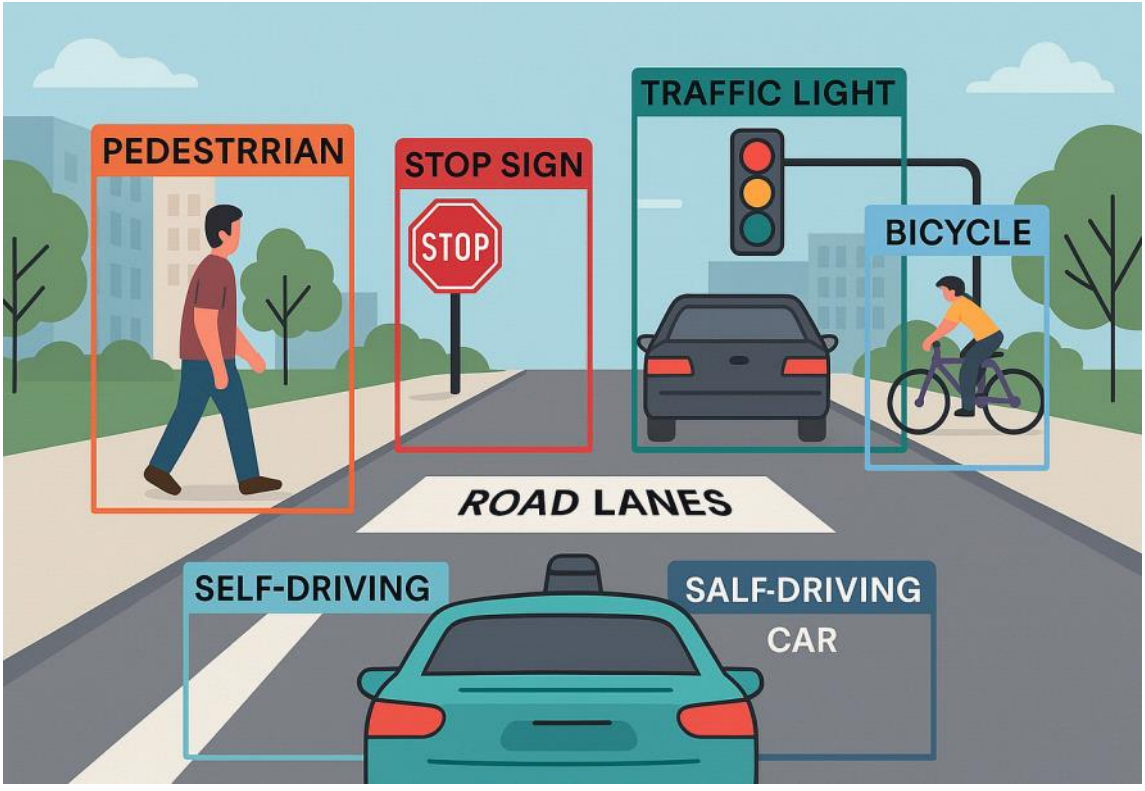
ANNOTATION

Data annotation:

Data annotation is the process of ***labeling data*** *whether it's text, images, audio, or video so machines can understand and learn from it.*
Imagine teaching a child to identify a dog. You'd show pictures and say, "This is a dog." Do that enough times, and the child begins to recognize dogs in new pictures without your help. That's exactly what data annotation does for AI.
In technical terms, annotation involves adding metadata (like tags, descriptions, or categories) to raw data so **machine learning models** can use it during training to recognize patterns, make decisions, or predict outcomes.
Real-Life Example: Self-Driving Cars
Self-driving cars are trained using millions of images and video frames. Each frame is **manually annotated** to label:



Pedestrians
Stop signs

Road lanes
Traffic lights
Vehicles
A data annotator will draw boxes around each object and label them ("car," "bicycle," "pedestrian"). These labeled images are then used to train the car's vision system.

Why Is Data Annotation So Important in AI/ML?

AI doesn't "know" anything on its own. It learns by example, **from annotated data**.
In **supervised learning**, a type of machine learning, data annotation is essential.
Without labeled examples, the algorithm cannot understand what it's trying to identify, classify, or predict.
Whether it's teaching an AI to recognize cats in photos or detect spam emails, **annotated data is the fuel** that powers AI training.